

Top IoT Trends Captured at eRocks

The highlight of this year's Electronics Rocks (popularly known as eRocks), held at Bengaluru in October 2013, was a dedicated track on the Internet of Things (IoT). Several experts presented insightful talks, from which the following key trends could be identified...

The Internet of Smartphones: Jayraj Ugarkar, head-Internet of Things CoE, Infosys Labs, spoke about how the IoT could help address the challenges faced by smartphones, and vice versa. Smartphones with the ability to act as sensors, accelerometers, gyroscopes, cameras and remote controls, and being something that people carry around all the time, could improve the adoption of IoT technologies, while the IoT in turn would result in expanding the scope of what smartphones offer.

Wi-Fi as an IoT interface: Earlier, Wi-Fi was not considered a common connectivity option in embedded systems due to the cost and size of embedding the required hardware in chips. However, today this is feasible. Hence, Jithu Niruthambath, director at Calixto Systems Pvt Ltd, suggested that engineers should seriously consider Wi-Fi as the first choice for an IoT interface, due to the extant availability of access points and the resulting ability to piggyback on the existing infrastructure.

Modern 'app' thinking needed in IoT products: As software becomes 'consumerware' and gets into a huge range of devices, developers must keep several golden rules in mind. Shivesh Vishwanathan, senior consultant (Mobility)-Technology Consulting Group, Persistent Systems, spoke about how developers must be aware of the backend functionality, integration, cloud-based platforms, service-based models, social connections, etc, in order to create the required experience on the client. They need to understand where the value resides in the product they are building, think very differently about how to slice and dice the functionalities for the end user, infuse their product with creativity; and aim for perfection even if it

means throwing away prototypes!

Rapid uptake of Bluetooth Smart: Nitin Gupta, lead engineer, µEnergy Applications, CSR India Pvt Ltd, discussed the reasons behind the popularity of Bluetooth Smart, the low-power variant of Bluetooth, and its features. Bluetooth Smart offers rapid rollout of low-power consumer products around the latest smartphone and tablet platforms, creating a large addressable market.

Gamering cross-domain expertise: Bringing the IoT to life requires a comprehensive approach that looks at sensing technology, intelligent embedded processing, connectivity, software and services in a holistic manner. Understandably, Narang N Kishor, mentor and principal design architect of Namix Technolabs, recommended that people and companies aiming to succeed in the IoT era must garner 'cross-domain' expertise, as well as skills in open and interoperable standards, protocols and interfaces, security, privacy and socio-ethical issues.

IoT in automobiles: Automobiles, especially electric vehicles, might well be one of the most active elements in the IoT scene, as they promise to be highly connected and combine a horde of infotainment technologies for the user's comfort. These include navigation, in-car communication, car-to-car communication, and integration with the user's phone for entertainment and communication.

Growing popularity of open source platforms: Open source hardware and software platforms, like Android and Contiki, are becoming very popular in the IoT. And, with the availability of Yun, Arduino's new Wi-Fi enabled flavour, things are expected to get even better. ●

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A smart smoke alarm

Nest Protect is a smart smoke alarm. Well, as you might have guessed, it contains dozens of sensors, is network-enabled, and has all the IoT functions. Let us tell you



about one really street smart feature in this alarm! Generally, people experimenting with deep-fried food, barbecues or grills tend to switch off the fire alarm because it will start buzzing. And they forget to switch it on again. Nest Protect has a feature that allows you to simply wave at a buzzing alarm to shut off the sound, in case there is no real fire, and the alarm then keeps quiet for a while. You do not have to switch it off, in case you forget to switch it on again!

A 'connected' camera

Stockholm-based Q Lifestyle has come up with a connected camera called theQ. And, it is more connected than just being Bluetooth or Wi-Fi enabled. theQ uses a 3G SIM card to directly upload all your photos from the camera to the Web, wherever you are. This product is for people who prefer a full-featured camera over a mobile phone, but do not like to lug along their cables and laptops wherever they go.

One app to rule over them all

If you have a dozen smart devices, and need to have a separate app to control each of these, then those devices are not so smart after all! OpenRemote, an open source software platform for connecting Internet-connected devices, hopes to change this, by enabling you to control all kinds of smart home devices regardless of the manufacturer. You can connect complex devices like smart air-conditioners and TVs as well as LED lights or simple devices made smart using DIY IoT platforms. However, at the moment, the tool is not targeted at common consumers, and can be used only by developers and those with a technical bent of mind as it requires a lot of programming. Over time, it is likely to become more user friendly, though currently, it is still pretty powerful if you can program it.

Read more about the Internet of Things at internetofthings.electronicshub.com